Physics
Slightly above standard level
Thermodynamics Practice Paper

Diploma Programme
Programme du diplome
Programa del Diploma

The last day summative, very official so has no errors in tasks. At all. 100%.

Candidate session number ☐☐☐☐☐☐☐☐

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estimated time 20-200 min

Instructions to candidates

- Write your session number in the boxes above.
- Answer all questions.
- For Section A, select one answer for each multiple-choice question.
- For Section B, answers should be written in the spaces provided.
- A calculator is required for this paper.
- A clean copy of the physics data booklet is required for this paper.

Section A: Multiple Choice

Answer all 16 questions. Select one answer for each question.

1. A heat engine wastes $4000\ \mathrm{J}$ of energy during a cycle. The work done by the heat engine during the same cycle is $1000\ \mathrm{J}$.

What is the heat engine's efficiency?

A. 0.20

B. 0.25

C. 0.75

D. 0.80

2. For a Carnot engine with an efficiency of 0.50 , $T_\mathrm{c} = 100\ \mathrm{K}$, where T_c is the temperature at which heat leaves the engine.

What is T_h , the temperature at which heat is transferred into the engine?

A. $400\ \mathrm{K}$

B. $200\ \mathrm{K}$

C. $100\ \mathrm{K}$

D. $50\ \mathrm{K}$

3. The first law of thermodynamics

A. involves the concept of entropy.

B. states that the internal energy of an ideal gas consists only of potential energy.

C. is the law of conservation of energy applied to thermodynamics.

D. claims that when a gas expands adiabatically, its temperature increases.

4. Which of the following statements is not an expression of the second law of thermodynamics?

A. Energy does not move spontaneously from a cooler to a hotter object.

B. It is impossible to convert all the energy from a hot object directly into work without energy losses.

C. During an irreversible change, the entropy of an isolated system always increases.

D. The total energy of an isolated system remains constant over time.

5. What happens to entropy during a reversible and during an irreversible process?

	Entropy during a reversible process	**Entropy during an irreversible process**
A	Unchanged	Unchanged
B	Unchanged	Increases
C	Increases	Unchanged
D	Increases	Increases

6. Liquid nitrogen in an open container at its boiling point $(-196\ ^\circ\mathrm{C})$ is placed in a sealed room and is allowed to completely evaporate.

Consider the following statements about the process:

I. The entropy of the nitrogen molecules decreased.

II. The entropy of the room decreased.

III. The entropy of the universe decreased.

Which of these statements are incorrect?

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

7. What is true about a closed system and an isolated system?

A. Energy can enter a closed system, but not an isolated system.
B. Mass can enter a closed system, but not an isolated system.
C. Energy can enter an isolated system, but not a closed system.
D. Mass can enter an isolated system, but not a closed system.

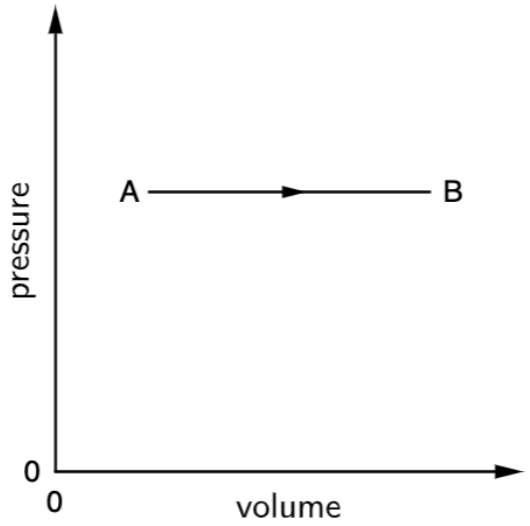
8. An air conditioner is installed inside a sealed container.

What is true about the entropy of the container?

A. It decreases.
B. It increases.
C. It does not change.
D. It is zero.

9. The graph shows how the pressure of an ideal gas varies with its volume during change AB.

Change AB is



A. adiabatic.

B. isothermal.

C. isovolumetric.

D. isobaric.

10. A functioning freezer, with its door open, is placed inside a thermally isolated room.

What happens to the temperature of the room and to the entropy of the room?

Temperature of the room

Entropy of the room

A.

Increases

Increases

B.

Decreases

Increases

C.

Increases

A. Decreases

B. D.

C. Decreases

D. Decreases

11. The reservoirs of a Carnot engine are maintained at temperatures of $200\ °C$ and $300\ °C$, respectively.

What is the best estimate for the efficiency of the engine?

A. 0.17

B. 0.33

C. 0.67

D. 0.83

12. Three statements about entropy are:

I. The entropy of a non-isolated system can locally decrease.

II. The change in the entropy of a system is inversely proportional to the absolute temperature at which heat is transferred to the system during a reversible process.

III. The entropy of gas molecules in a sealed container is proportional to the number of possible microstates of the molecules.

Which of these statements are correct?

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

13. A gas undergoes a complete thermodynamic cycle. The internal energy of the gas before the cycle is E_1 and after the cycle it is E_2. What is true about E_1 and E_2?

A. The relationship between E_1 and E_2 can only be determined if we know the initial volume, pressure, and temperature of the gas.

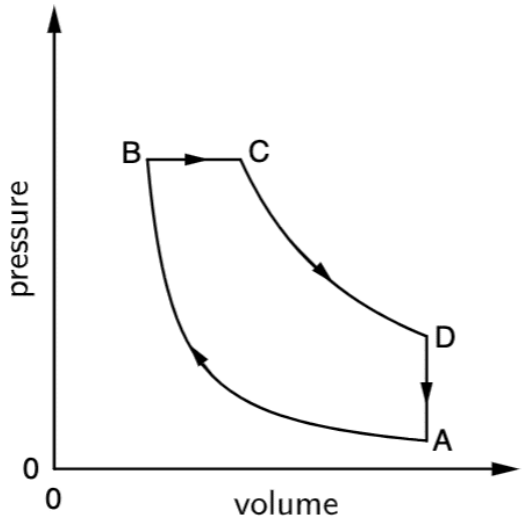
B. $E_1 > E_2$

C. $E_1 < E_2$

D. $E_1 = E_2$

14. The diagram shows cycle ABCDA for an ideal gas.

Work is done by the gas during



A. changes BC and CD.

B. changes DA and AB.

C. changes BC and DA.

D. changes AB and CD.

15. During an experiment a monatomic ideal gas is subjected to an adiabatic compression, transitioning from initial state Y to a final state with volume V . The work done on the gas during the process is W . What is the work done on the gas and the change in the internal energy of the gas during an isothermal compression from initial state Y to a final state with volume V ?

	Work done	**Change in internal energy**
A	Greater than W	Less than zero
B	Less than W	Greater than zero
C	Greater than W	Zero
D	Less than W	Zero

16. An ideal gas is used to operate a Carnot engine. The energy transferred from the gas to the cold reservoir per cycle is $6000\ \mathrm{J}$ and the energy absorbed by the gas from the hot reservoir per cycle is $10000\ \mathrm{J}$. The cold reservoir is kept at a constant temperature of $27\ ^\circ\mathrm{C}$ and the hot reservoir is kept at a constant temperature of $T\ ^\circ\mathrm{C}$.

What is T ?

A. 45

B. 68

C. 230

D. 480

1. Answer all questions. Answers must be written within the answer boxes provided.

A container is filled with an ideal, monatomic gas and sealed with a movable piston. The volume of the gas is 3.7 m^3, the temperature of the gas is 28 °C and the pressure of the gas is 7.2 Pa.

a. Outline the difference between the internal energy of an ideal gas and the internal energy of a real gas.

[2]

b. Find the number of gas atoms in the container.

[2]

The piston on the container is moved so that the volume of the container decreases by 67 %. The temperature of the gas is kept unchanged.

c. Calculate the new pressure of the gas.

[1]

d. Explain, in terms of molecular motion, this change in pressure.

[2]

(a) [2]

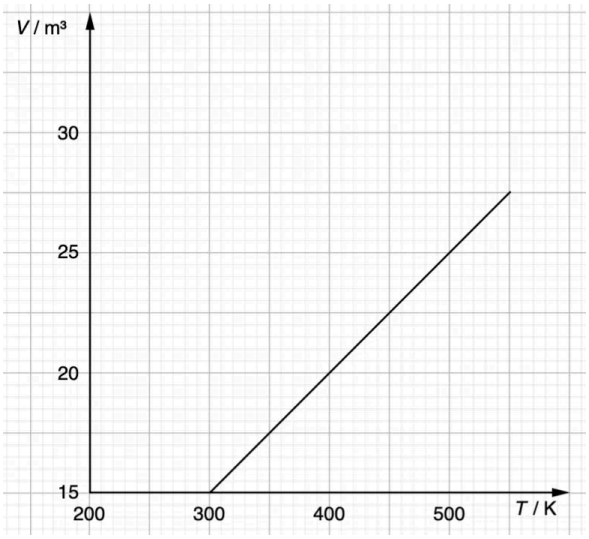
(b) [2]

(c) [1]

(d) [2]

2. Answer all questions. Answers must be written within the answer boxes provided.

An expandable balloon is filled with a fixed mass of argon-40 gas. The pressure of the gas is maintained at 1.3 kPa. The graph shows how the volume of the gas varies with temperature.



a. State two assumptions of the kinetic model of an ideal gas.

[2]

b. Explain why argon-40 behaves as an ideal gas for temperatures between 20 °C and 70 °C.

[2]

c. Find the mass of the argon-40 gas in the balloon.

[2]

(a) [2]

(b) [2]

(c) [2]